

Trade Study TS-003

Ground Control Station Enclosure Selection

Project: Instrument Case Modernization for GCS as a subsystem of TOMTOM

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1. Objective

The objective of this trade study is to evaluate candidate enclosure architectures for the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM) Ground Control Station (GCS) and select the enclosure that best satisfies the project's operational, technical, and cost requirements. The enclosure serves as the structural foundation of the Ground Control Station and is responsible for protecting internal electronics while supporting portable field deployment, subsystem integration, maintenance, and future hardware expansion.

Because the enclosure directly affects system portability, durability, maintainability, and internal packaging, the selected design must provide adequate internal volume for the computer hardware, power distribution system, communications equipment, cooling components, and future mission hardware while remaining suitable for repeated field use.

2. Design Requirements

Requirement	Value
Portable	Required
Rugged Construction	Required
Internal Electronics Protection	Required
Internal Expansion Space	Required
Easily Modified	Required
Carry Handles	Required
Weather Resistance	Desired
Budget	<\$250 Preferred
Field Maintainable	Desired

The Ground Control Station enclosure must provide a rugged and portable platform capable of protecting internal electronic equipment while supporting field deployment, transportation, maintenance, and future subsystem expansion. Because the enclosure functions as the primary structural component of the Ground Control Station, it must provide sufficient internal space to accommodate the repurposed ASUS G73S computer hardware, power distribution system, communications equipment, cooling components, and future mission hardware without compromising accessibility or maintainability.

The enclosure must also support structural modifications necessary for prototype development, including installation of custom mounting brackets, display supports, cable routing, ventilation, power interfaces, and removable equipment trays. In addition to meeting immediate prototype requirements, the selected enclosure should preserve flexibility for future hardware upgrades while remaining cost-effective and suitable for repeated operational use.

3. Alternatives Considered

Alternative A

Pelican Equipment Case

Pelican equipment cases are widely used throughout military, aerospace, and industrial applications because of their exceptional durability and environmental protection. These injection-molded cases provide excellent resistance to impact, moisture, and dust while offering numerous interior customization options. Their widespread use has established them as the industry benchmark for rugged portable equipment enclosures.

Advantages

- Extremely rugged construction
- Waterproof and dust resistant
- Excellent reputation for reliability
- Large aftermarket accessory support
- Available in numerous sizes

Disadvantages

- Highest acquisition cost
- Limited internal mounting features
- Additional modification required for hinged workstation configuration

Alternative B

Pro Cainz Portable Instrument Case

The Pro Cainz portable instrument case is a steel field equipment enclosure originally designed for transporting instrumentation and electronic equipment. The enclosure provides a large internal workspace, removable lid, and robust mechanical construction that make it well suited for conversion into a portable Ground Control Station. The existing military-inspired appearance also aligns with the intended operational role of the TOMTOM system.

Advantages

- Excellent internal volume
- Strong steel construction
- Easily modified
- Low acquisition cost
- Suitable for custom workstation configuration
- Supports future subsystem expansion

Disadvantages

- Heavier than polymer equipment cases
- Not factory waterproof
- Requires modification to install hinges, electronics, and internal mounting hardware

Alternative C**Apache Protective Case**

Apache protective cases provide many of the benefits associated with premium rugged equipment cases while offering significantly lower acquisition cost. These cases include foam interiors and weather-resistant construction suitable for protecting electronic equipment during transportation.

Advantages

- Lower cost than Pelican
- Weather resistant
- Lightweight
- Readily available

Disadvantages

- Less durable than premium equipment cases
- Limited internal working depth
- Reduced internal mounting flexibility

Alternative D**Custom Aluminum Enclosure**

A fully custom aluminum enclosure provides maximum design flexibility by allowing every structural component to be designed specifically for the TOMTOM Ground

Control Station. While offering complete customization, this approach significantly increases fabrication complexity, manufacturing cost, and project schedule.

Advantages

- Fully customizable
- Lightweight construction
- Excellent future expandability
- Professional appearance

Disadvantages

- Highest fabrication effort
- Increased manufacturing cost
- Longer development time
- Specialized fabrication equipment required

4. Evaluation Criteria

Criterion	Weight
Cost	20%
Durability	20%
Ease of Modification	20%
Internal Capacity	15%
Portability	10%
Expandability	10%
Weather Resistance	5%

The candidate enclosure architectures were evaluated using a weighted decision matrix based on structural performance, lifecycle considerations, and project constraints established for the TOMTOM Ground Control Station. Evaluation criteria were selected to reflect the operational requirements identified during system design while balancing technical capability, long-term maintainability, and overall project cost.

Durability and ease of modification received the highest weighting because the enclosure must withstand repeated transportation and field deployment while accommodating extensive structural modifications throughout prototype development. Cost was also assigned a significant weighting to ensure the selected architecture remained compatible with available project resources without sacrificing critical functionality. Internal capacity and expandability were evaluated to ensure sufficient space for current hardware while preserving flexibility for future subsystem integration. Portability and weather resistance were included to assess the enclosure's suitability for field operations and its ability to protect sensitive electronic equipment under varying environmental conditions.

5. Decision Matrix

Criteria	Weight	Pelican	Pro Cainz	Apache	Custom Aluminum
Cost	0.20	2	5	4	1
Durability	0.20	5	5	4	4
Ease of Modification	0.20	3	5	3	5
Internal Capacity	0.15	4	5	3	5
Portability	0.10	4	3	5	4
Expandability	0.10	4	5	3	5
Weather Resistance	0.05	5	3	4	3

Each alternative was evaluated using a qualitative scoring scale ranging from one to five, where a score of one represents poor performance relative to the evaluation criterion and a score of five represents excellent performance. Individual criterion scores were assigned using engineering judgment, published product specifications, anticipated modification effort, expected lifecycle performance, and compatibility with the operational requirements established for the TOMTOM Ground Control Station.

The weighted decision matrix provides a structured and repeatable methodology for comparing multiple enclosure architectures while reducing subjective bias during the engineering decision-making process. By multiplying each criterion score by its associated weighting factor, the matrix emphasizes characteristics considered most critical to successful system performance while allowing lower-priority considerations to contribute appropriately to the final evaluation. This approach improves decision traceability and provides quantitative justification for selecting the preferred enclosure architecture.

6. Weighted Scores

Alternative	Total Score
Pelican	3.70
Pro Cainz	4.70
Apache	3.70
Custom Aluminum	3.80

Weighted scores were calculated by multiplying the score assigned to each evaluation criterion by its corresponding weighting factor and summing the resulting values for each candidate enclosure. The resulting total score represents the overall suitability of each alternative relative to the established project requirements.

Although the weighted score serves as the primary decision metric, the final engineering decision also considers practical implementation factors that cannot be fully represented by numerical scoring alone. These factors include hardware availability, expected fabrication effort, compatibility with the Ground Control Station architecture, future subsystem expansion, and overall project objectives. Consequently, the selected enclosure represents the solution that provides the best overall balance between technical capability, lifecycle value, implementation effort, and long-term system flexibility rather than simply the highest numerical score.

7. Discussion

The Pro Cainz portable instrument case achieved the highest weighted score among the evaluated alternatives and was selected as the preferred enclosure architecture for the TOMTOM Ground Control Station. Although heavier than polymer protective cases, the enclosure provides the best overall balance of durability, internal capacity, ease of modification, expandability, and acquisition cost while supporting the operational objectives of the project.

One of the primary advantages of the selected enclosure is its generous internal volume, which provides sufficient space for integration of the repurposed ASUS G73S computer hardware, power distribution system, battery, display assembly, cooling components, communications equipment, and future subsystem expansion. The steel construction also provides a rigid structural platform capable of supporting custom mounting brackets, hinged display mechanisms, removable equipment trays, cable management hardware, and other modifications necessary to convert the enclosure into a fully integrated Ground Control Station.

Unlike commercial protective cases designed primarily for equipment transportation, the Pro Cainz enclosure offers greater flexibility for structural modification and long-term development. Although premium enclosures such as Pelican cases provide superior environmental protection, their higher acquisition cost and reduced adaptability for extensive customization reduce their overall value within the project's current constraints. Likewise, while custom aluminum enclosures offer maximum design flexibility, the increased fabrication effort, manufacturing complexity, and development time are not justified for the current prototype phase.

The selected enclosure establishes a robust mechanical foundation that supports the current Ground Control Station architecture while preserving sufficient flexibility for future hardware upgrades, additional mission equipment, and continued system development throughout the TOMTOM project.

8. Decision

Selected Alternative:

Pro Cainz Portable Instrument Case

Rationale

The Pro Cainz enclosure provides the best balance between:

- Structural durability
- Internal capacity
- Ease of modification
- Expandability
- Cost
- Long-term maintainability

The selected enclosure establishes a robust structural foundation for the TOMTOM Ground Control Station while preserving sufficient flexibility for future hardware integration, subsystem expansion, and continued prototype development.